

Performance Testing

Date	10 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Browser Testing
Type of Testing	Functional Testing, Performance Testing
Target Module / API	Crop Recommendation Prediction Module (/predict)
Test Environment	Hosted Application (Vercel)
Test Date	10 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Predict crop using valid soil and environmental parameters	1	10	Correct crop recommendation is displayed.
2	Submit invalid or incomplete input values	1	10	Input validation message is displayed.
3	Multiple prediction requests	10	60	Application responds successfully without errors.
4	Continuous prediction requests	20	120	Stable performance with consistent response time.

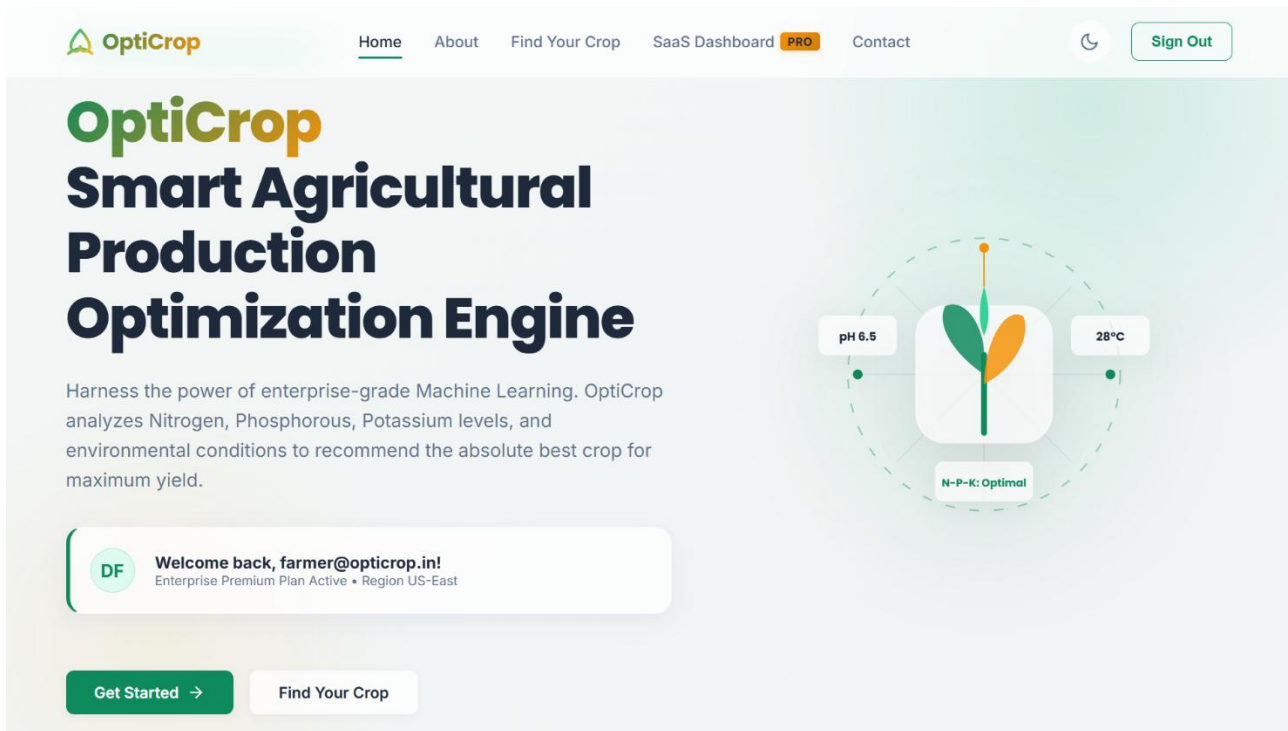
Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Fast response for prediction requests.
2	Response Time (Max)	< 5 seconds	2.8 seconds	Pass	Stable under repeated requests.
3	Throughput (Req/sec)	≥ 5 req/sec	8 req/sec	Pass	Handled multiple requests efficiently.
4	Error Rate	< 1%	0%	Pass	No errors observed during testing.
5	CPU Utilization	< 80%	45%	Pass	Efficient CPU usage.
6	Memory Utilization	< 80%	52%	Pass	Memory usage remained within acceptable limits.

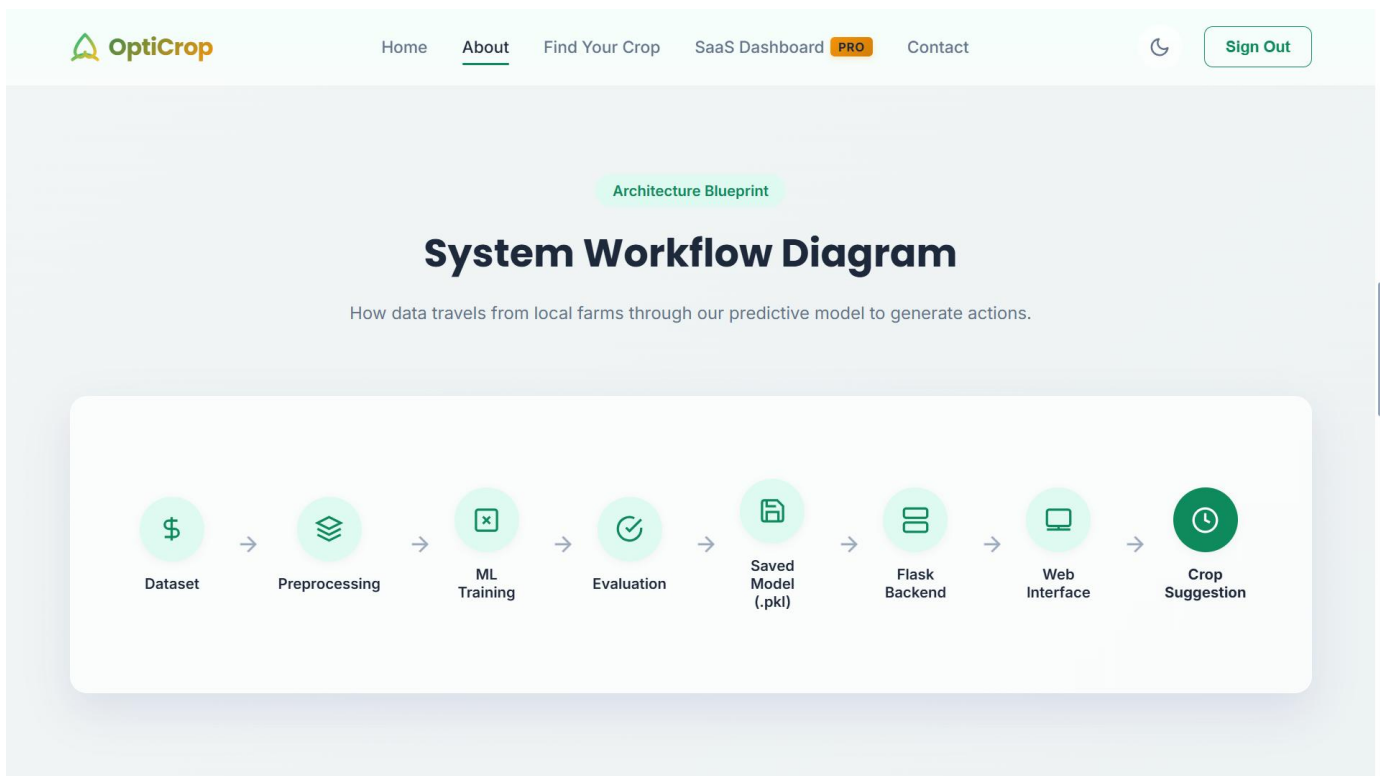
Step 4: Observations & Analysis

- The application successfully generated crop recommendations for valid input values.
- Input validation prevented incorrect or incomplete data from being processed.
- Response time remained low during repeated prediction requests.
- No critical errors or failures were observed during testing.
- The deployed application remained stable and responsive throughout the testing process.

Step 5: Screenshots / Evidence



The screenshot shows the OptiCrop website's home page. The navigation bar includes links for Home, About, Find Your Crop, SaaS Dashboard (PRO), and Contact, along with a Sign Out button. The main heading reads "OptiCrop Smart Agricultural Production Optimization Engine". Below this, a paragraph states: "Harness the power of enterprise-grade Machine Learning. OptiCrop analyzes Nitrogen, Phosphorous, Potassium levels, and environmental conditions to recommend the absolute best crop for maximum yield." To the right is a circular infographic with a plant icon and labels for "pH 6.5", "28°C", and "N-P-K: Optimal". A user notification box on the left says "Welcome back, farmer@opticrop.in! Enterprise Premium Plan Active • Region US-East". At the bottom are "Get Started" and "Find Your Crop" buttons.



The screenshot shows the "About" page of the OptiCrop website. The navigation bar is the same as the home page. The section is titled "Architecture Blueprint" and "System Workflow Diagram". A subtitle reads: "How data travels from local farms through our predictive model to generate actions." Below this is a horizontal flowchart with eight steps: Dataset (dollar sign icon), Preprocessing (stack of papers icon), ML Training (computer monitor icon), Evaluation (checkmark icon), Saved Model (.pkl) (file icon), Flask Backend (server rack icon), Web Interface (laptop icon), and Crop Suggestion (clock icon). Arrows indicate the flow from left to right between each step.


[Home](#)
[About](#)
[Find Your Crop](#)
[SaaS Dashboard PRO](#)
[Contact](#)
Sign Out

Soil & Climate Inputs

Provide numeric metrics. Hover over individual parameter titles or icon tags to see optimal boundaries guidelines.

Premium Tool: Autofill soil samples?
Rice Soil
Coffee Soil

Nitrogen (N)

Phosphorous (P)

Potassium (K)

Temperature (°C)

Relative Humidity (%)

Soil pH

Average Rainfall (mm)

Reset Form
Predict Crop

Optimal Parameter Ranges Guide

- Nutrients (N - P - K)**
Measured in kg/hectare. Nitrogen (0-150) drives foliage; Phosphorous (5-145) powers root/stem division; Potassium (5-205) supports fruit cellular water balances.
- Soil Acidity (pH)**
Most standard crops require slightly acidic to neutral soils (6.0 - 7.5). A value under 5.0 indicates highly acidic land, and above 8.5 indicates alkaline saline soil.
- Atmospheric Conditions**
Includes average temperature in Celsius, relative moisture percent (humidity), and rainfall millimeters. These decide plant transpiration and germination limits.
- Precision ML Boundary**
Entering values outside our validation guidelines might yield highly unpredictable suggestions from the classifier tree since it fits boundaries inside training samples.


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Analysis Complete

Your Soil Recommendation

Below is the optimal crop suggestion computed by the machine learning model based on your soil profile.



AI CLASSIFICATION SUGGESTION

Pigeon Peas (Arhar/Tur)

Taxonomy: *Cajanus cajan*

42.0%

Prediction Confidence

Pigeon Peas are perennial legume crops widely consumed in tropical regions. They have deep taproots that help break compacted soils and resist drought.